

General Knowledge, Computer & Indian Important Events

Questions and Answers Compilation

Section 1: General Knowledge

Q: What is the capital of India?

Ans: New Delhi is the capital of India. It serves as the seat of all three branches of the Government of India and houses important institutions such as the Rashtrapati Bhavan, Parliament House, and the Supreme Court. New Delhi was formally inaugurated as the capital in 1931, replacing Calcutta (now Kolkata), which had been the capital during British colonial rule.

Q: Which is the longest river in the world?

Ans: The Nile River, flowing through northeastern Africa, is traditionally considered the longest river in the world, stretching approximately 6,650 kilometers. It passes through eleven countries including Egypt, Sudan, and Uganda, and has historically been vital for agriculture, transportation, and the rise of ancient civilizations such as Egypt, which depended on its annual floods for fertile soil.

Q: What is the smallest planet in our solar system?

Ans: Mercury is the smallest planet in our solar system, with a diameter of about 4,879 kilometers, making it even smaller than some moons like Ganymede and Titan. It is also the closest planet to the Sun and has extreme temperature variations, ranging from scorching heat during the day to freezing cold at night, due to its thin atmosphere.

Q: Who wrote the national anthem of India?

Ans: Rabindranath Tagore, the renowned Bengali poet and Nobel laureate, wrote 'Jana Gana Mana,' which was adopted as the national anthem of India. He composed the original song in Bengali in 1911, and it was later translated and adapted. Tagore was the first Asian to win the Nobel Prize in Literature, awarded to him in 1913 for his collection of poems titled Gitanjali.

Q: Which gas is most abundant in Earth's atmosphere?

Ans: Nitrogen is the most abundant gas in Earth's atmosphere, making up about 78 percent of its composition. Oxygen follows at roughly 21 percent, with the remaining 1 percent consisting of argon, carbon dioxide, and trace gases. Nitrogen is essential for plant growth and is fixed into usable forms by certain bacteria and through industrial processes like the Haber process.

Q: What is the largest mammal on Earth?

Ans: The blue whale is the largest mammal, and indeed the largest animal, ever known to have existed on Earth. It can grow up to 30 meters in length and weigh as much as 200 tons. Despite its massive size, the blue whale primarily feeds on tiny shrimp-like animals called krill, consuming several tons of them daily during feeding seasons.

Q: Who is known as the Father of Modern Physics?

Ans: Albert Einstein is widely regarded as the Father of Modern Physics due to his groundbreaking contributions, including the theory of relativity and the famous equation $E=mc^2$. His work fundamentally changed our understanding of space, time, and energy. Einstein was awarded the Nobel Prize in Physics in 1921, primarily for his explanation of the photoelectric effect.

Q: Which country is known as the Land of the Rising Sun?

Ans: Japan is known as the Land of the Rising Sun because, from the perspective of China and other Asian countries, the sun appears to rise from the direction of Japan in the east. This name is reflected in the Japanese flag, which features a red circle representing the sun on a white background, symbolizing the nation's identity and history.

Q: Which is the largest desert in the world?

Ans: The Antarctic Desert is technically the largest desert in the world, covering about 14 million square kilometers, since deserts are defined by low precipitation rather than temperature. If considering only hot deserts, the Sahara in Africa is the largest, spanning roughly 9.2 million square kilometers across North Africa and featuring extreme temperatures and vast sand dunes.

Q: Who discovered gravity?

Ans: Sir Isaac Newton is credited with formulating the law of universal gravitation after observing a falling apple, according to popular legend, in the late 17th century. His work, published in Principia Mathematica in 1687, explained how every object in the universe attracts every other object, laying the foundation for classical mechanics and influencing centuries of scientific thought.

Q: What is the chemical symbol for gold?

Ans: The chemical symbol for gold is Au, derived from the Latin word 'aurum,' meaning shining dawn. Gold is a dense, soft, and highly malleable metal known for its resistance to corrosion and tarnishing, which has made it valuable throughout human history for jewelry, currency, and various industrial and electronic applications.

Section 2: Computer Knowledge

Q: What does CPU stand for and what is its function?

Ans: CPU stands for Central Processing Unit, often referred to as the brain of the computer. It is responsible for executing instructions from programs by performing arithmetic, logical, control, and input/output operations. Modern CPUs contain multiple cores that allow them to process several tasks simultaneously, significantly improving the speed and efficiency of computing devices.

Q: What is the difference between RAM and ROM?

Ans: RAM, or Random Access Memory, is a volatile type of memory used to store data temporarily while a computer is running, and it loses its contents when the power is turned off. ROM, or Read-Only Memory, is non-volatile and retains data even without power, typically used to store firmware or instructions needed to start up a computer, such as the BIOS.

Q: What is an operating system?

Ans: An operating system is software that manages computer hardware and software resources while providing common services for computer programs. It acts as an intermediary between users and the computer hardware, handling tasks such as memory management, process scheduling, and file systems. Popular examples include Microsoft Windows, macOS, Linux, and Android.

Q: What is the full form of HTTP?

Ans: HTTP stands for HyperText Transfer Protocol, which is the foundation of data communication on the World Wide Web. It defines how messages are formatted and transmitted between web browsers and servers, and how servers and browsers should respond to various commands. A secure version, HTTPS, encrypts this data using SSL or TLS protocols for safer browsing.

Q: What is the difference between a virus and a worm in computing?

Ans: A computer virus is malicious software that attaches itself to a legitimate program or file and requires human action, such as opening an infected file, to spread and cause damage. A worm, on the other hand, is a standalone malicious program that can replicate itself and spread across networks without needing to attach to another program or requiring user interaction.

Q: What is cloud computing?

Ans: Cloud computing refers to the delivery of computing services such as servers, storage, databases, networking, and software over the internet, commonly known as 'the cloud.' This model allows businesses and individuals to access resources on demand without owning physical infrastructure, offering benefits like scalability, cost efficiency, and flexibility. Examples include Amazon Web Services, Microsoft Azure, and Google Cloud.

Q: What is the function of a compiler?

Ans: A compiler is a special program that translates source code written in a high-level programming language into machine code or an intermediate form that a computer's processor

can execute directly. Unlike an interpreter, which translates and executes code line by line, a compiler processes the entire program at once, generally resulting in faster execution of the final program.

Q: What is the difference between LAN and WAN?

Ans: A LAN, or Local Area Network, connects computers and devices within a limited geographical area such as a home, office, or school, allowing fast data transfer rates. A WAN, or Wide Area Network, spans much larger geographical distances, often connecting multiple LANs across cities, countries, or continents, with the internet itself being the largest example of a WAN.

Q: What is the difference between hardware and software?

Ans: Hardware refers to the physical, tangible components of a computer system, such as the motherboard, hard drive, keyboard, and monitor, which can be touched and seen. Software, on the other hand, consists of the intangible instructions and programs that tell the hardware what to do, including operating systems, applications, and utilities, none of which have a physical form.

Q: What is an IP address?

Ans: An IP address, or Internet Protocol address, is a unique numerical label assigned to every device connected to a computer network that uses the Internet Protocol for communication. It allows devices to locate and communicate with each other across networks. IP addresses come in two main versions, IPv4, which uses a 32-bit format, and the newer IPv6, which uses a 128-bit format to accommodate more devices.

Q: What is artificial intelligence?

Ans: Artificial intelligence refers to the simulation of human intelligence processes by computer systems, including learning, reasoning, problem-solving, and decision-making. It encompasses various subfields such as machine learning, natural language processing, and computer vision, and is increasingly used in applications ranging from virtual assistants and recommendation systems to autonomous vehicles and medical diagnosis.

Section 3: Important Events in Indian History

Q: When did India gain independence from British rule?

Ans: India gained independence from British rule on August 15, 1947, after a long struggle led by leaders such as Mahatma Gandhi, Jawaharlal Nehru, and many others through movements of non-violent civil disobedience. The partition of British India also occurred at this time, leading to the creation of Pakistan, which resulted in widespread migration and communal violence affecting millions of people.

Q: What was the Quit India Movement?

Ans: The Quit India Movement was launched by Mahatma Gandhi on August 8, 1942, demanding an end to British rule in India. It was a mass civil disobedience movement that called for immediate independence and saw widespread participation across the country, despite the British government's harsh crackdown, including the arrest of major leaders of the Indian National Congress.

Q: When was the Indian Constitution adopted?

Ans: The Indian Constitution was adopted by the Constituent Assembly on November 26, 1949, and it came into effect on January 26, 1950, a date now celebrated annually as Republic Day. Dr. B. R. Ambedkar is widely regarded as the chief architect of the Constitution, having chaired the Drafting Committee responsible for its formulation.

Q: What was the Jallianwala Bagh massacre?

Ans: The Jallianwala Bagh massacre occurred on April 13, 1919, in Amritsar, Punjab, when British troops under General Reginald Dyer opened fire on a large peaceful crowd gathered to protest against repressive colonial policies. Hundreds of people were killed and many more injured, and the incident became a turning point that intensified the Indian independence movement against British rule.

Q: What is the significance of the Dandi March?

Ans: The Dandi March, also known as the Salt March, was led by Mahatma Gandhi from March 12 to April 6, 1930, covering about 390 kilometers from Sabarmati Ashram to the coastal village of Dandi. It was a non-violent protest against the British salt tax and monopoly, and it sparked widespread civil disobedience across India, drawing international attention to the independence movement.

Q: When did India become a republic?

Ans: India became a republic on January 26, 1950, when its Constitution came into full effect, replacing the Government of India Act of 1935 as the country's governing document. This date was chosen to commemorate the Declaration of Purna Swaraj, or complete independence, made by the Indian National Congress in 1930, and it is celebrated annually as Republic Day.

Q: What was Operation Blue Star?

Ans: Operation Blue Star was a military operation conducted by the Indian Army in June 1984, ordered by Prime Minister Indira Gandhi, to remove armed militants from the Golden Temple

complex in Amritsar, Punjab. The operation resulted in significant casualties and damage to the sacred site, and it led to widespread political and communal tensions, eventually contributing to the assassination of Indira Gandhi later that year.

Q: What was the significance of India's first general election?

Ans: India's first general election was held between 1951 and 1952, marking the country's transition into a full-fledged democracy under universal adult suffrage. It was one of the largest democratic exercises in history at the time, with millions of citizens, including a vast number of illiterate voters, participating to elect representatives to the first Lok Sabha and various state assemblies.

Q: When did India conduct its first nuclear test?

Ans: India conducted its first nuclear test, codenamed Smiling Buddha, on May 18, 1974, at the Pokhran test range in Rajasthan, making it the sixth country in the world to successfully detonate a nuclear device. This was followed by a series of additional tests in May 1998, known as Pokhran-II, which firmly established India's status as a nuclear weapons state.

Q: Who was the first Prime Minister of India?

Ans: Jawaharlal Nehru served as the first Prime Minister of independent India, taking office on August 15, 1947, and continuing in the role until his death in May 1964. Known for his vision of a secular, democratic, and industrially developing nation, Nehru played a key role in shaping India's foreign policy, including the Non-Aligned Movement during the Cold War era.

Q: What was the Battle of Plassey and its significance?

Ans: The Battle of Plassey, fought on June 23, 1757, was a decisive conflict between the British East India Company, led by Robert Clive, and the Nawab of Bengal, Siraj-ud-Daulah. The British victory, aided by the betrayal of Mir Jafar, marked the beginning of British colonial expansion in India and laid the foundation for nearly two centuries of British dominance over the subcontinent.

Section 4: Indian Geography and Polity

Q: Which is the highest mountain peak in India?

Ans: Kangchenjunga, located on the border of Sikkim and Nepal, is the highest peak located within Indian territory, standing at approximately 8,586 meters above sea level, making it the third-highest mountain in the world. It is considered sacred by local communities, and climbing expeditions traditionally stop short of the actual summit out of respect for these beliefs.

Q: How many states and union territories does India have?

Ans: India currently consists of 28 states and 8 union territories, each with its own administrative structure, though union territories are generally governed more directly by the central government compared to states, which have their own elected legislatures. This number has changed over the years due to reorganizations, such as the formation of Telangana in 2014 and changes to Jammu and Kashmir in 2019.

Q: What is the role of the President of India?

Ans: The President of India serves as the ceremonial head of state and the formal head of the executive, legislature, and judiciary, though most executive powers are exercised by the Prime Minister and the Council of Ministers in practice. The President is elected indirectly by an electoral college consisting of members of Parliament and state legislatures, serving a term of five years.

Q: What is the significance of Article 370?

Ans: Article 370 of the Indian Constitution granted special autonomous status to the state of Jammu and Kashmir, allowing it to have its own constitution and significant legislative independence from the rest of India, except in matters of defense, foreign affairs, and communications. This provision was revoked by the Indian government in August 2019, leading to the reorganization of the region into two union territories.

Q: What is the Sabarmati Ashram famous for?

Ans: The Sabarmati Ashram, located near Ahmedabad in Gujarat, was established by Mahatma Gandhi in 1917 and served as his residence and the headquarters of his movement for Indian independence for many years. It was from this ashram that Gandhi launched the famous Dandi March in 1930, and today it stands as a museum and memorial dedicated to his life and teachings.

Section 5: Sports, Awards and Miscellaneous

Q: What is the Bharat Ratna?

Ans: The Bharat Ratna is the highest civilian award in India, instituted in 1954 to recognize exceptional service toward the advancement of art, literature, science, and public service. Recipients have included figures such as Dr. B. R. Ambedkar, Mother Teresa, and cricketer Sachin Tendulkar, who in 2014 became the youngest person and the first sports person to receive this honor.

Q: When did India win its first Cricket World Cup?

Ans: India won its first Cricket World Cup in 1983, defeating the West Indies in the final at Lord's Cricket Ground in London under the captaincy of Kapil Dev. This victory is widely regarded as a turning point for cricket's popularity in India, inspiring a generation of players and dramatically increasing public interest in the sport across the country.

Q: What is ISRO and what is it known for?

Ans: ISRO, the Indian Space Research Organisation, is India's national space agency, established in 1969 and responsible for the country's space program, including satellite launches and planetary exploration. It gained international recognition for missions such as Chandrayaan, India's lunar exploration program, and Mangalyaan, which made India the first Asian nation to reach Mars orbit successfully in 2014.

Q: What is the significance of the Green Revolution in India?

Ans: The Green Revolution refers to a period beginning in the 1960s during which India adopted modern agricultural techniques, high-yielding crop varieties, and increased use of fertilizers and irrigation to boost food production significantly. Spearheaded by agricultural scientist M. S. Swaminathan, this transformation helped India move from a nation dependent on food imports to one largely self-sufficient in grain production.

Q: Who was awarded the first Nobel Prize from India?

Ans: Rabindranath Tagore became the first Indian, and the first Asian, to receive a Nobel Prize when he was awarded the Nobel Prize in Literature in 1913 for his collection of poems titled Gitanjali. Since then, other Indians, including physicist C. V. Raman and economist Amartya Sen, have also received Nobel Prizes in various fields, contributing to India's global intellectual recognition.